

ActInf Textbook Group ~ Cohort 3 ~ Meeting 14 (Chapter 6, part 1)

TextbookGroup/ParrPezzuloFriston2022/Cohort_3 | Cohort 3 ~ Meeting 14 (Chapter 6, part 1) | 58:13

hello everyone it's June 13th 23

and we're in cohort three

discussing chapter six

so we'll head over to the chapter six

page

and uh before we look at some of the

specific questions and like pick ones

that are interesting and so on

does anyone just want to give any

overall thoughts or questions

okay

well of course just write in the

chatter raise your hand

if you want

um we can

look to some of the previously addressed

uh or at least asked questions

and get some more perspectives and

improve some of the discourse

does anybody have a alternative

preference

all right so also definitely please vote

with a interesting button

because once we get above like 10 or or

some number of uh

votes then we'll make a short video so

we'll kind of be upvoting and then

seeking to improve those sections most

all right

why is modeling of the generative

process a necessary step for building an

active inference model

who wants to give a first thought on

this

that we can review but first just yet

Ali and then anyone else just why

why are we talking about modeling the generative process

uh yeah I think as we talked about uh the the distinction between General to model and generative process before uh well at least uh according to the previous active inference framework generative process is

uh basically what

um I mean it refers to the latent state of uh the observatory data we get from the environment uh in other words uh what causes those uh data uh in this case uh is put into a kind of probabilistic model so

on the other hand generative model

I mean the aim of generative model is to provide a kind of mechanism to track those uh generative process as close as it can or at least

based on the situation of Interest so that's one distinction between General to model and generative process but on the other hand

um well recently I

um got to the understanding that it's not so straightforward to

separate generative model from General to process as because in some situations we might never know what generative process uh

is I mean what it actually constitutes generates a process and the only thing we can concentrate or we can at least uh confidently focus on is to model the generative uh model based on uh some of the

assumptions about how the environment
will

behave in the future or at least based
on some of our knowledge about the
overall mechanism of the environment
so but in some simple situations it
might be possible to

differentiate between the two

awesome thank you

anyone else have like any related
thought or question

about the role of building a generative
process in a generative model

and then we'll look at some of the
previous responses

and then we'll move on to another
question

okay I'll bring in one um paper path
integrals particular kinds and of course
remember that you can always just click
on the on the the icon of the institute
on the right and you'll jump to where we
are

um so in this paper which we often look
back towards

we see this

taxonomy of different kinds of things
ranging from the simplest inert systems
through active systems

but perhaps simple active systems like a
metronome

on through sophisticated or strange
systems

which have hidden internal states that
are doing like counterfactual kind of
world modeling self in World modeling
so in chapter six

we're talking about whether you're

trying to set up a simple inert system
or a simple octave system or you're on
the path towards making more complex
cognitive models
there's going to be certain steps that
are outlined in the recipe
so that is like the continuity of active
inference type modeling of the
particular partition
splitting things into internal external
and blanket States and then further
differentiating the blanket into the
incoming sensory States and the outgoing
action States
we're calling them sense and action is
of course
conditioned upon choosing one side of
that blanket to be internal and that's
kind of like the cognitive system of
interest and so we call the cognitive
system of interest in terms of its
particular States the internal and the
blanket States
that is like the thing
and that is the generative model
Ali

I get and one other related thing is I
mean this typology is actually based on
a three-level classification uh namely
at one hand we have the distinction
between active and inert particles which
do and do not have active States and
then on the other level we have
distinction with between dissipative and
conservative particles which are and are
not subject to random fluctuations and
finally the ordinary and strange
particles

um which have active states which do you
have and do not have active States and
one other point is it's interesting that
in some recent literature especially uh
the one the white paper published
um about four months ago ecosystems
designing ecosystems of intelligence
from first principles and this recent
paper

the inner screen model of Consciousness
the term generative process never appear
even once in the entire paper so
basically uh they they've began begun to
use a generative model interchangeably
with generative process in some sense
yes

um this is this is something that we're
gonna be
asking about

in guest stream 45 in six days
because we noticed that in this
presentation that Maxwell had generative
model covering the entire
particular partition

rather than the way that it was shown
in um the figures of the textbook
where we had the generative model
with internal and blanket States
essentially being partitioned from the
generative process which is the process
which actually gives rise to the
observations

so
we can think of it as kind of like some
uh
dialectical usage

in the dialect sense of the active
ontology so

Evolution and use of what people are referring to

um

that's how it goes when things are changing fast but we'll find out in a few days some more takes on this

um

active inference agent employs a generative model as its model the external world

and uses the generative model to perform inferences about perceptual input to make decisions about actions

the external world or the generative

process is complex and unknowable

generative models may or may not be able to learn

but learning and also action selection

puts the generative model in better

alignment with the world's generative

process whatever it might be two ways to

minimize free energy change your mind

through perception and learning change

the world through action

and there's some more

discussion any other

for anyone

a comment or a question about the

generative process and the generative model

and kind of where we're at in chapter

six or we'll move on to the next

question

all right awesome and also please use

the upvote so that we can just

figure out which questions we want to

make improvements and um short videos

around

okay

next question again please just raise

your hand if you have if you have a

different question or put it in the chat

so this is kind of a core

question about chapter six

what are the four steps in the recipe to

construct an active inference model

so

from the very first sentence of chapter

six

what the chapter is aiming to provide is

this four-step recipe to construct an

active inference model now it doesn't

have to be strictly linearly followed

they're more like four considerations

or four areas of questioning more so

than like you must take on one two three

four in this order

um

but

um let's look at the Active inference

model recipe we made just a sub page

because this is such a central topic

I mean all of our expectations and

preferences slash hopes and dreams

about applying active inference

using active inference modeling for some

system or phenomena of Interest

this is

kind of what it comes down to again this

isn't the only or the total

protocol for modeling but chapter six is

the

transition point in the book

where the first five chapters set us up

to understand epistemically what active

inference models were about

the low road and the high road in
chapters two and three how we approach
active inference
chapter 4 the generative model itself
and some of the mathematics and the
formalisms of it and then chapter five
exploring some applications in mammalian
neurophysiology and now chapter six is
like

our system of Interest whether it's
another aspect of mammalian
neurophysiology or whether it's any
other of the many many situations that
we're interested in modeling this is the
model recipe as laid out by the authors
here so does anyone want to give a
thought or a question or just overview
on the four

considerations that that are in chapter
six for the model recipe

yeah Ali and then anyone else

yeah just one thing uh that I think uh

We've also talked about before is to

keep in mind that it's not absolutely

necessary uh to follow those steps uh

strictly in a sequential order so it

would be possible to uh to address them

in at least uh in uh

an asynchronous way or even uh in some

cases even change the order of the steps

so it's just a general framework to

follow the recipe it's not I think needs

to be strictly observed

yeah maybe it's more like making four

dishes

but let's look at what they are

so first question which system are we

modeling

and this is
analogous to the identification of the
system of interest in systems
engineering
and there are situations where this is
trivial
or where the teacher tells you which
system is being modeled
but there's also situations where it's
non-trivial
because of course everything is attached
to everything in the world and so making
these distinctions about the spatial and
the temporal and the causal boundaries
of the system that we're actually going
to choose to model
such that it's at some base optimal
trade-off point between being tractable
and understandable implementable all of
these different features that's kind of
iterated process and that's not one
that's just simply written on the world
as it is so we just get to show up and
find out like this is where we have to
take an active approach to modeling to
even determine some of these processes
through inquiry but what which system
are we modeling
the um second question is what is the
most appropriate form for the generative
model so let's look at where they
describe this in a bit more detail
foreign
this is the decision whether it is
appropriate to think of a process in
terms of categorical or discrete
inferences
or continuous inferences

motivating the choice between discrete or continuous time implementations or hybrid of active inference so just to recall figure 4.3 is where we see a juxtaposition of the discrete time active inference generative model on the top and the continuous time active inference generative model on the bottom they're being laid out this way to highlight their structural similarities like they have this kind of downward facing e and then the upstairs with action in both cases but there are some differences between the discrete time and the continuous time formulation we're not going to go into too much more detail on their differences right now but we can discuss if people want and this is going to get explored in a lot more detail in chapters 7 and 8 the coming two chapters which are explicitly about discrete and continuous time generated models and you can also make hybrid implementations so the most common hybrid implementation is um in the folk psychological setting where decision making is discrete at a higher level like should I um reach for the the orange or the Apple and then at the sensory motor or lower smaller or faster level you have a continuous time model like a motor reflex model and then the motor reflex model the attractor of where it's going to be

expecting to find itself which is where
it reaches towards
is being imposed from the top down by
that categorical decision-making model
so Hybrid models are absolutely possible
and one setting that they're used in is
systems that have discrete
decision making processes that interface
with continuous sensory motor activities
then we need to select the most
appropriate hierarchical depth
motivating the choice between shallow
versus deep models
so
how much
spatial temporal or causal depth is the
model biting off
are you trying to model leather for one
season or for one year or for 20 years
or for a million years
and then how is that scope
going to be
compartmentalized into the model so
let's just say we wanted to do an 100
Year predictive model
one approach would be
um one tick of the model
one time step
in the discrete time model is a hundred
years and we fit all the parameters so
that they are like what's accurate after
100 years go by
but then we wouldn't be able to
necessarily make a prediction at 50
years so okay what else can we do we
could have the tick be one year and have
this model go 100 into the future
or we could have a nested model where we

have decades where the tick goes by one year at a time and then 10 decades and so at the higher nested level it's like the transition between decades and so these are all structural variants of a model that can do prediction of 100 years or maybe you want every single day maybe you want a transition Matrix amongst seconds but each of these are going to have trade-offs in terms of the computational requirements of the model whether you're going to be able to calibrate it with data the amount of parameters that you're going to be creating and so this is the modeler's quest and Art and Science to find the right models for a given situation because you know all models are wrong some are useful and finally we need to consider whether it's necessary to endow the generative model with temporal depth kind of previously mentioned and the ability to predict action contingent observations to support planning that's called planning as inference and it is using the expected free energy or if you have different actions that can be taken different policies which is kind of like an action over the given time Horizon then planning as inference means predicting the consequences of those actions and then doing a type of inference to determine which path of action to be

taken

after determining the form for the

generative model

there are the questions of setting up

the generative model and setting up the

generative process

so

the kernel of active inference if we see

it in figure 4.3 again the discrete time

or the continuous time

this doesn't have

attention metacognition

um

learning of preferences

counterfactual World modeling what would

happen if they did that

there's a lot of sophisticated cognitive

phenomena that are not found in the base

model in fact the base model doesn't

even have learning at all

we're going to see in the coming

chapters how this can be

straightforwardly elaborated to include

learning on any parameter shown here

but this by itself does not include

learning

and so they're they're highlighting

that there's a lot of decisions to be

made

in terms of setting the priors

and which aspects are going to be fixed

and what aspects are going to be

learnable and remember that the models

are maps not territories so just because

something is learnable

in principle or in practice in the real

world

doesn't necessarily mean that the model

is going to be simply mimicking that
and vice versa
so again these are just some steps to
consider
essentially setting up
the generative model and the generative
process
the the type of thing that we're
modeling the cognitive entity and
their environment
and they say that these four steps in
most cases suffice to design active
inference model
once completed the behavior of the
system is determined by the standard
schemes of active inference
this is a huge advantage of working with
active inference cognitive modeling
is once you get the model set up
and entered into whatever implementation
approach you're using SPM or Pi mdp or
RX infer Etc
then
the methods of parameterizing it
fine-tuning it fitting that model
are very generic and straightforward
so it's not like you kind of slap
together how you're thinking about the
system and then you're going to spend
all of this engineering work on making
the optimization
tractable
it's the opposite
a lot of the work goes into
a structural understanding of the system
because once you have your model entered
in
to Pi mdp for example

then standard routines are used to fit the model

Ali than anyone else

uh yeah I'm wondering uh how I mean to what extent uh this recipe is applicable when it comes to modeling the collective intelligence which is uh I mean the current interests of active inference research largely uh and in particular I believe based on the stages of development proactive inference outlined in designing ecosystems white paper currently we're under systemic intelligence level of modeling and we're aiming toward modeling or toward developing an efficient way to model a sentient intelligence but we're not there yet

on the uh problem of modeling sophisticated intelligence and sympathetic and ultimately shared intelligence so it's interesting to see how those four steps can be applied when when we finally get into those stages of development of active impress agents yes this is from the paper from 2022.

Olivia

yeah oh so I suppose that you know the best way to understand these four different points is to uh look at how you know uh current examples or papers uh have done them uh so take you know already implemented models uh and identify you know uh step one two three and four um and you know do we have in uh in the active Enterprise textbook groups uh

some kind of uh you know list of
um example models to do that yes
yes absolutely so it's a it's a great
suggestion though I agree that um
showing these stages being carried out
we can hope becomes
a practice in the active inference
ecosystem because it would connect a lot
of the dots for people replicating the
analyzes and for learners but here's
what we can look at now in the code page
we have several dozen
different languages different
implementations of active inference
drawn from a variety of uh papers pretty
much the GitHub pulled out where we can
now not all these are heavily documented
or pedagogically oriented
so sometimes there's some forensic work
to be done with like understanding
um because and also a lot of this
happens outside of the context of shared
epistemic niches and shared education so
the variables are going to have
different names and the implementations
are going to be quite different these
are not necessarily created
um after reading chapter six especially
if they yeah but but it certainly is a a
retroactive approach that can describe
um and then I think one of the most
clearly unpacked and didactic examples
that we can look at is uh model stream

7.2

and in model stream 7.2 in the
video description is The Notebook
of this
presentation

and so it does in a much more
pedagogical way and in an executable way
it runs perfectly as is
you can see
how is B getting set up
so again this isn't exactly aligned with
the four stages of the recipe but but
like we said those don't have to have in
that order but this is like these are
the variables that you need to set up a
b c d e
yeah and then at the bottom you see the
routines
the Run active inference Loop
so if seeing it in a pseudo code way
helps you understand the perceptual
nature of the active inference Loop
better than the analytical equations
that's definitely a good way to learn
and study this
okay yeah great
any more that we can say on this
one other resource I'll link in there
this is to the octave block friends
um Coda this is another project at The
Institute so here we we've kind of
expanded on the recipe
to include
more up front and along the way
so more of like the structural
understanding of the system and why the
model is being made then
we kind of include the steps of chapter
six
but also take more of a life cycle
perspective on the operations of these
models
and then we have all these different

settings where people have started to explore those kinds of modeling approaches so I'll just link to this page also um in in the recipe okay let's go back to the questions so what are the four steps we just looked through them essentially they're a set of considerations that build the generative model the generative process so the cognitive thing and the environment the niche that's giving observations for considerations that help us develop that from its kind of sketchiest outlines on through the determination of which variables are going to be like learnable or not which ones are discrete State space which one's our continuous State space Etc and then once we have that set up and entered into our implementation approach then all the routines are standard and there's always exciting developments with making those routines better at dealing with sophisticated um planning and so on so again this is a big strength and a growing one for active inference that the work in understanding the system gets us to the point where it can be run generically as opposed to just taking an ad hoc understanding of a system which is like kind of a mistake but one that flies by

in many situations
and then confusing the engineering work
with improving the computational aspects
of the model
with an understanding of the system
okay
so that was this question
again upvote the question so they rank
higher but I'm just going to continue
down from the top
um in figure 6.1
what do the unidirectional and
bi-directional arrows mean
so we see there are some connections
between nodes nodes are random variables
and edges represent causal influences
possible causal influences
just because they're here doesn't mean
that they're relevant in a given
situation
but this is kind of like the set of the
possible connectivities
of the action perception Loop
so
the blanket states are connected to each
other
possibly because they form that
insulating set around the internal
States
so again you can design a system where
the API call comes in and then a second
computer computes it and then a third
computer emits it
so there doesn't have to be an edge
between these two
but in principle it's allowable
um
we see this symmetry

where again an internal and external
that just depends which system we're
focusing on but there's a symmetry
across here so you can kind of see it's
like a rotated 180 degree symmetry where
sensory states don't have a backwards

Arrow from internal States

and external states don't have a
backwards Arrow to active States

however external and sensory have
bi-directionality and internal action
have bi-directionality so

um

the particle

that which is being partitioned off from
the world

can have different

sensory and active blanket States

blanket being the joint set of U and Y
and internal States

and these arrows are just shaping out
the space of the possible interactions
amongst those variables

and

um in

let's see guest stream

number seven

how particular is the physics of the
free energy principle

if people want to go into more detail on
this question

uh Miguel Aguilera and colleagues

provided some technical analyzes and

basically asked under what topologies of
the action perception Loop

does the free energy principle apply

for example if we have like a simple
around-the-clock model

none of these backwards arrows do things
still apply
or if we have more or less connectivity
in in principle what applies
and um the short answer to this is
basically it's always going to hold in
some fashion but it may or may not be
interesting
and the important constraints
that we're really adhering to
the most important constraint is
across the interface
there's no telepathy and there's no
telekinesis
that doesn't mean internal States
can't track or phase lock onto external
states with very sparse sensory input
the point is they do so only through the
blanket so you can't get information
about hidden States except by sensory
States that's no telepathy clause
in the no telekinesis Clause is you can
only influence external States
through the octave States the blanket
so the fact that there's no connection
between internal and external States is
really the most important constraints on
the topology of
perception and action and active
inference and other than that there's
some technical discussions to be had
about exactly
um which situations have bi-directional
arrows and so on and some of it is
explored in 7.1
Ali and then anyone else
I just want I wanted to mention a couple
of technical terms

um that being used in the literature repeatedly one is the autonomous states which refers to um The Joint set of internal States and the activist States instead of the whole blanket state so particle is the joint set of internal States and the whole blanket States but if we consider only the active States among the blanket States then that's been referred to autonomous States and the other term is a sparse coupling which exactly refers to that unidirectional Arrow in this figure so very briefly sparse coupling refers to the fact that things are defined by what they are not connected to so if the whole diagram here would be connected the whole states here would be connected through bi-directional errors then that would just that wouldn't be a sparse coupling so the reason this term is used is because of that unidirectionality in the speaker awesome and one important piece on the autonomous States and this is kind of like a cool way to think about active formally but also informally autonomous States again are just the internal and active States so these are the two ways that you can reduce your free energy change your mind change the world through action learning a perception update internal States action updating active States our preferences the C Vector are over

sensory States

so we say I prefer to find myself at 37c

homeostatic body temperature

and so because I prefer that I'm going

to seek policies that

Meander my way even if it's one step

backwards Two Step forwards

to reduce my surprise bound my surprise

reduce my expected free energy about

finding myself at homeostatic body

temperature so our preferences are over

sensory States

but the two things that we can control

the autonomous States

like all knowing and all-powerful

are the internal and active States

and another way of saying that is you

don't directly have your thumb on the

scale

we want to see a certain number on the

scale

but all we want to be able to do even in

our maximal empowerment would be perfect

updating of learning and perception and

perfect selection of action

if you enable the model to also take

control over its sensory incoming data

not the not the downstream processing of

it but the primary sensory data

then the model may ins in pursuit of

seeking its preferences just simply

modify the sensory data

so

if we'd like to see a sunset you know we

want to either learn but we're not

seeing one we want to either learn that

we actually don't prefer that

then all is good with the world or make

a plan to see it
but if you can just turn on the VR
in your mind and see it right there
you've kind of
um
made a hollow satisficing of your
preferences
and in the ecological evolutionary
context you're going to get swept off
the table
so the autonomous states are like what
we would want to control under maximal
empowerment and they're the two Avenues
of reducing free energy changing mind
and changing the world

Olivier

yeah I mean uh I'm not sure I got it uh
but you you said the preference is put
on the sensory uh state
correct yeah uh so
if that
um reflected in the mathematical uh
expression of free energy
oh absolutely it is not shown here this
on a particular partition we don't see
the C vector or preferences but um let's
look at it in the equations
so here equation 2.6
on expected free energy
and so
in the top decomposition
expected free energy G is a function
that's evaluating policies so it's
loading in our policy prior that's like
our habits
and then it's sharpening those policy
priors into a policy posterior
that then you can sample from or just

select the best option from
so let's just say that um two-thirds of
the time I turn left that's my habit
and then if we just passed the prior on
policy through then action would be
selected according to that ratio
two-thirds the time going to the left
however what expected free energy does
is it sharpens the policy prior habit
into policy posterior in the top
decomposition

we see one way of thinking about what
contributes to the expected free energy
for a given policy concern or option and
that's thinking about it in terms of the
information gain and the pragmatic value
so first The Information Gain here we
see that there's a KL Divergence between
hidden States Through Time X tilde
 P_i is on both sides and here there's a y
so if the Y the expected sensory
information under that policy if it
didn't give us any information it's not
expected to give us information then the
KL Divergence is zero

no information is expected to be gained
from that course of action however if
the policy does have a sensory
observations that would lead us to learn
then epistemic gain is non-zero
here we have pragmatic value
and this is basically how surprised are
you

about the sensory data given your
preference

and so if your preference is 37 and it's
like a bowl around 37 then the highest
pragmatic value policy

would be the least surprising meaning
that the ball would be right at the
bottom of the bowl at 37.
so just to contrast with reward learning
reinforcement learning
in reward and reinforcement learning you
have this reward function that's like 37
is the most rewarding body temperature
so I'm going to try to find my way up to
the most rewarding point
we take an opposite approach we say I
expect and prefer to be at 37c
that's what I'd be least surprised by if
my thermoreceptors were at 37.
that's why expectations and preferences
are kind of used together in that way
and then policies are selected
that are able to
um
contribute to pragmatic value and
Information Gain
and during the modeling process you see
that there are some fine tuning
needed to balance these two drives and
that's actually explored in this
notebook like if you set the preference
for getting sugar to 1 million to one
then it'll never seek information it's
ready to risk it for the chance of
winning a million but if the pragmatic
value is um very small then the
Information Gain term dominates
and another way that that is shown in a
figure
is in figure 2.6 where we have expected
free energy just as shown now here the
pragmatic value term is on the left but
it's the same equation yeah and so if

only pragmatic value matters so it's a
fully observable there's nothing
epistemic to gains like a chessboard
then you get expected utility Theory
in contrast when there's no pragmatic
value on the table so all outcomes are
equally pragmatically valuable then you
only have two three four five
in which case you're having information
driving decision making
yeah

but both can be on the table and that is
how octave inference in a first
principles way is negotiating the
explore exploit dilemma

yeah

thank you

Ali thanks

and also this kind of information
seeking or a preference seeking is what
referred to as

curiosity or rather artificial curiosity
in the literature but it's probably
worth mentioning another recent Trend in
active inference literature which is the
development of sophisticated active
inference

which I don't think is

explicitly stated in the textbook but
basically sophisticated inter inference
um

deals with the situations in which
rather than

trying to answer what will happen if I
do this they try to address what will I
believe if I do this so basically
it's it's based on the consequences of
actions for beliefs rather than just the

states of the world

thank you

yeah especially with all of the

developments and elaborations happening

like around the time of the writing of

this textbook and since then

um

it's kind of It kind of contextualizes

the bigger picture but like figure 4.3

is like the Lego block

this is like the the neutral minimal

cognitive kernel

shown in its discrete time and

continuous time setting

and a lot of the work right now on the

analytical and on the formal side for

example with category Theory

is to understand the compositionality of

these models like can we really attach

this to that

in principle

and then a lot of the computational work

which you'll see

um in the code section

are using techniques from computer

science broadly

like search on large trees and branching

and pruning on trees

using these computer science and

engineering approaches to make running

large active inference models more

tractable

so the textbook helps us um

parachute in

and get a first overview

but

for those who want to see more there's

certainly more on the computer science

implementation side and on the
analytical on the formal side
but all things considered the textbook
s the line
between that
and again chapter six is like the moment
in the book where it transitions from
the first five chapters
which are like a background and
statistics class
on through chapter six and Beyond which
through our work we can even develop
into something like a A playbook
or a set of code and and in cohort three
which we're in right now with chapter
six one of our goals for this activity
interval
is to leave an executable Trail
so we want to develop the answers
to questions
but also especially where we can to
um
leave a code
or executable or just pseudo code
way to approach this
and that's something that can be
collaboratively worked on inversion
through the cohorts
okay a few more questions
all right how does the particular
partition so figure 6.1
particular partition the particle
blanket and internal States autonomous
States just internal and active but the
particle imagine if you're looking at a
dust particle under the microscope
that would be an inert particle but a
cognitive particle is like the agent in

the agent-based modeling
so we see sometimes
a simple action perception Loop
and other times we see the the more um
expanded
particular physics
but then
this question asks
how does the particular partition relate
to the partially observable Markov
decision process in figure 4.3
so where did the Markov blanket go in
figure 4.3
okay one simple thought that anyone can
add more the observations here oh
are the sensory States
and so in the continuous time they're
even shown with Y so that's actually the
same variable as used for sensory States
and the particular partition
but the O are the observables
across the markov blanket and then here
 π 's policy selection policy inference
which is influencing the B Matrix which
is how the agent thinks things change
Through Time
but we could also Imagine That π also
reaches out to influence the generative
process which just isn't shown here
here just the agent's generative model
is shown
there's no generative process but if
there were a generative process it would
be
receiving an input from P_I
and sending an output to observations
so the pomdp can be seen as like a way
of unrolling

the particular partition

because this gives a um sequential

linear executable format

to modeling the particular partition

any other thoughts on this

Ali

uh yeah one other way to like look at it

is to interpret the uh I mean because in

sorry in pomdp uh the a the agents

preferred observations

um

somehow strives toward keeping it within

the high probability States uh which

would be consistent uh with its

continued existence in other words uh

this uh striving toward uh keeping these

hyper probability States can be

interpreted as keeping uh its markup

blankets marker blanket intact so in

other words markup blanket is kind of

implicitly

um implied in Palm pmdp but not uh

obviously explicitly stated but it can

be interpreted

from various perspectives

yes thank you

okay

gibsonian affordance concept

does anyone want to give a little

information on this question and then I

think we'll have a response

well it's a controversial issue indeed

yeah if you can just briefly uh yeah

remind uh Gibson's affordance concept

yes

okay yes let me get the

right page here

okay here I'm going to

um

share some work by Paulo an intern

and he's analyzing

the affordance concept and the

representation Concept in the English

and in the Portuguese literature using

the active inference ontology to to

connect between the two of them

um and there's a table

so recently there was a um paper written

by ramstead

about the affordance 3.0 concept

and basically says the way that we talk

about affordance is an active that is

pretty much the gibsonian affordance

concept

so as of the

um tactfully noted this is like an area

of discussion slash controversy so I

look forward to Apollo developing the

work more and and bringing this out

however

affordance 1.0 concept by Gibson

was that the affordance was about the

perception of a capacity for action in

the environment like a handle is

perceived as for grasping

and people focus a lot on this direct

perception and there's the whole Markov

blanket trick which we've talked about

elsewhere and people focus on like

whether the Marco blanket is direct or

indirect perception but that is actually

a little bit of a red herring

the key piece of the gibsonian

affordance concept is that the

affordance is doing something

explanatory about perception it's

explaining a perception of a certain type which is the perception for action in the niche

however inactive inference as it's been used

affordance

is used to describe action capacities that are not necessarily perceived

more like something that the system can do

and so when we look at the π variable from like figure 4.3

then like we call those a for what we call E affordances and then π is the policies of affordances so if it's just one time step then π is the same as e but if it's uh 12 time steps then it's all the 12-step combinations of the affordances

e what can be done

however active inference does not highlight the perceptual aspects of the affordance

because it's kind of a view from the inside view from the outside which we're going to come back to in chapter nine but just to show what this looks like the experimenter might say well that Mouse can go up down left right so that's how the active inference affordance concept is used up down left right are the four capacities for action

however the traditional ecological psychological

gibsonian affordance concept highlighted the view from the inside which is the perception of an action capacity

now that's not a perfect Concept in and
of itself because it just begets the
next question well what raises
Salient affordances to awareness
so it's not like one is better than the
other
and surely they can be understood of as
compatible
however there is some significant work
needed to understand the continuity of
these Concepts
and whether or not it's valid to say
that the affordance 3.0 concept is
actually in the direct lineage of the
affordance 1.0 and 2.0 concept
and Paulo s is the best intern to speak
to about this topic
cool
well
that brings us to the um
end here
I'll unlike these ones
for uh next week will be at the other
time
and um yeah everyone's welcome to add or
upvote questions
or anything else any anyone else want to
add anything before we close
awesome thank you yes chapter six is
when it really starts getting into the
the modeling and the doing
so also next week I'll I'll hope that we
can
um
develop out some of the code maybe even
annotate some of the
um the pi MTP examples or some of the
other examples in terms of the recipe

okay

thank you all farewell

thank you

thanks bye